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PCCO403 : Natural Language Processing

Title - Depression & Emotion Detection From Text

Group ID : 5

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ABSTRACT

Abstract: -

Depression and emotional distress can often be reflected through a person's choice of words, writing style, and emotional expressions in text. This project proposes an NLP-based system for detecting depression-related emotions from textual data such as social media posts, messages, or user-provided statements. The system will preprocess the text by removing noise, handling stop words, tokenizing sentences, and converting words into meaningful numerical features using techniques such as TF-IDF and word embeddings. A machine learning or deep learning classification model will then analyze the extracted features and classify the text into suitable emotional or depression-related categories. The system can provide an early indication of concerning emotional patterns and may support mental-health awareness and further professional assessment. It is intended as a screening and educational tool, not as a medical diagnosis.

Expected Input: -

A text statement, post, review, or message entered by the user, for example: "I feel hopeless and have lost interest in the things I used to enjoy."

Expected Output: -

The system will display the predicted emotional category, such as Positive, Neutral, Sad, Anxious, or Depression-related, along with a confidence score. Example: "Prediction: Depression-related emotion | Confidence: 87%."

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CHAPTER 1: INTRODUCTION

1.1 Background and Problem Statement

People express feelings in social posts, messages and other written text. Manually examining large text collections is time-consuming and subjective. NLP text classification can learn patterns from labelled examples and assign a text to a predefined category. This project studies two related but separate tasks: detecting one of six emotions and classifying text as depression-related or non-depression-related.

1.2 Objectives

- Build a six-class emotion classifier.
- Build a binary depression-related text classifier.
- Compare Logistic Regression, Multinomial Naive Bayes and Linear SVM using TF-IDF features.
- Evaluate the selected models on held-out test sets.
- Expose both classifiers through a Streamlit application.

1.3 Scope and Responsible Use

The project is limited to English text and the labels represented in the datasets. It does not estimate depression severity, assess suicide risk or provide clinical diagnosis. Predictions describe patterns in a text sample, not the mental-health status of its author.

CHAPTER 2: RELATED WORK

Traditional text classification commonly represents documents with weighted word features and trains a classifier on labelled examples. TF-IDF is a widely used weighting method for information retrieval and classification [1], [2]. Linear Support Vector Machines are effective baselines for high-dimensional sparse text features [3], [5], while Naive Bayes and Logistic Regression provide useful comparison models [4], [7].

Research on depression-related language and social-media signals highlights challenges involving weak or subjective labels, privacy, domain shift and the difference between text patterns and clinical assessment [11]–[14]. These concerns motivate cautious use of the current system. The project is an applied comparison of classical methods rather than a claim of clinical or state-of-the-art performance.

2.1 Research Gap and Project Approach

The project brings two text-classification tasks into one application and compares the same three families of classical classifiers. Separate models and TF-IDF vectorizers are retained for each task because their label sets, training data and vocabularies differ.

CHAPTER 3: SYSTEM DESIGN

A user submits one text sample through the Streamlit interface. The text is preprocessed and passed through two independent pipelines. Each pipeline transforms text with its fitted TF-IDF vectorizer, then uses its saved Linear SVM model to produce a separate prediction.

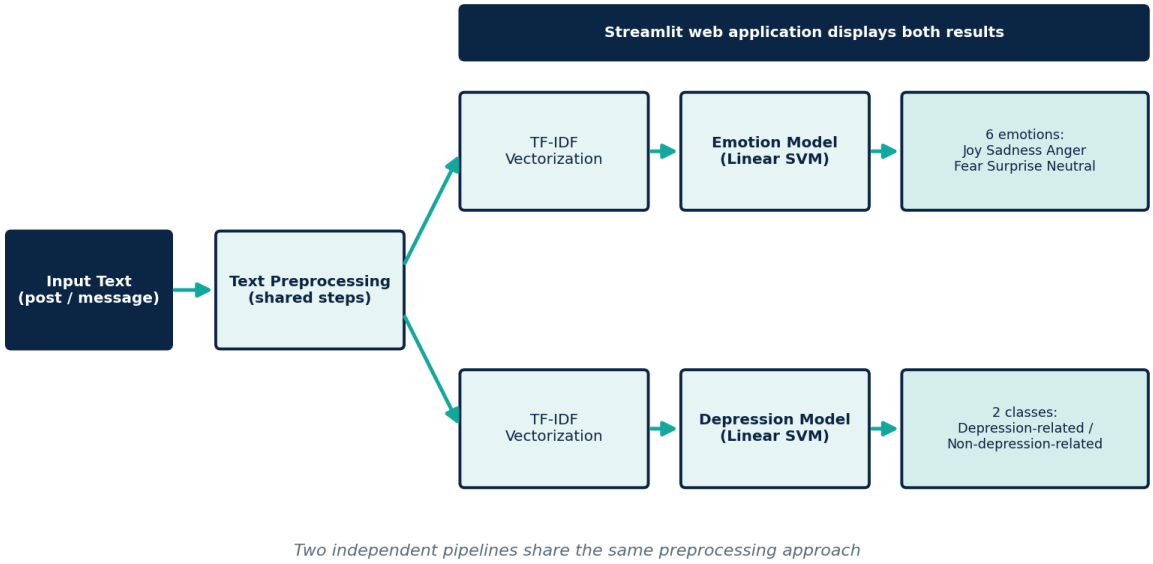


Figure 1. System architecture showing independent emotion and depression-related pipelines.

3.1 Main Functional Requirements

ID	Requirement
FR1	Accept text input from the user.
FR2	Clean and preprocess text consistently with training.
FR3	Generate TF-IDF features with the fitted vectorizers.
FR4	Predict one of six emotion labels.
FR5	Predict depression-related or non-depression-related text.
FR6	Display both results in the Streamlit interface.

3.2 Runtime Workflow

1. User enters text.
2. Text is cleaned and tokenized; stopwords are filtered while negation words are retained.
3. Each fitted TF-IDF vectorizer converts the text into numeric features.
4. The two saved Linear SVM models predict independently.
5. The interface displays the emotion and depression-related text classification.

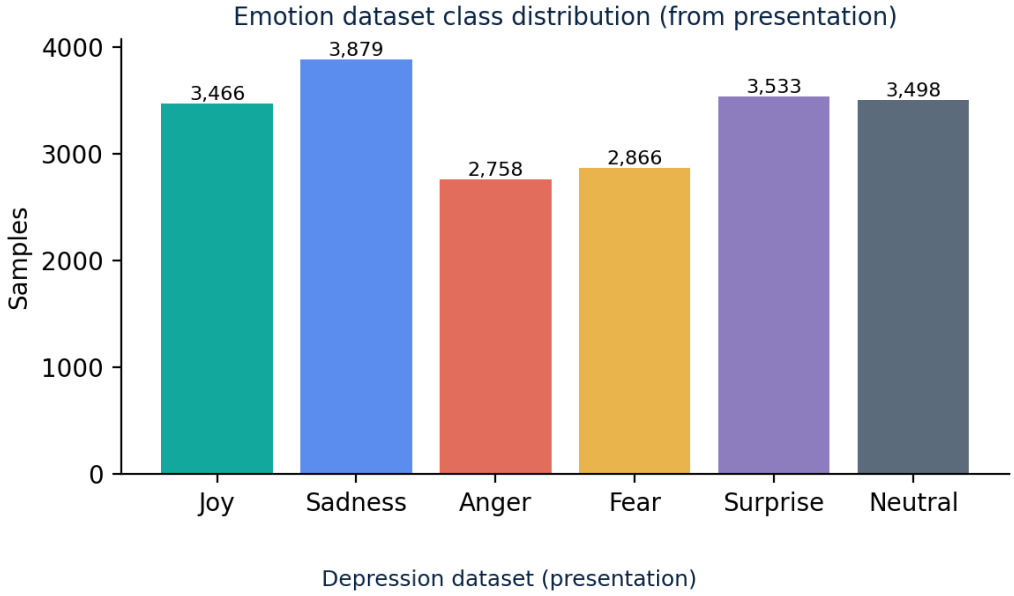
CHAPTER 4: DATASET AND METHODOLOGY

4.1 Dataset Summary

The project uses two labelled datasets. The emotion dataset has 20,000 examples divided into six classes. The depression-related dataset has 200,000 examples split evenly between two labels. Dataset labels and source distributions are inherited from the datasets; balanced class counts do not guarantee that real-world text has the same distribution.

Dataset	Total	Labels	Train / validation / test
Emotion	20,000	Sadness, Joy, Love, Anger, Fear, Surprise	16,000 / 2,000 / 2,000
Depression-related text	200,000	Non-depression-related, Depression-related	112,000 / 28,000 / 60,000 for model evaluation

Figure 2. Emotion dataset class distribution.



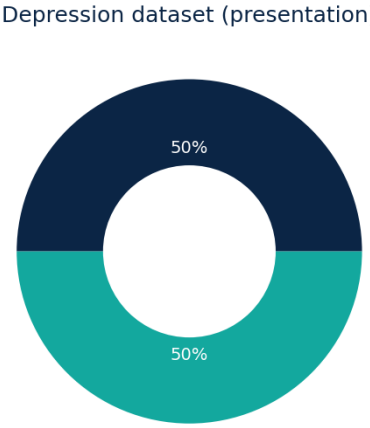


Figure 3. Depression dataset class distribution.

4.2 Text Preprocessing

Preprocessing reduces noise and makes text suitable for vectorization. The implemented steps include normalizing text to lowercase, removing URLs and punctuation noise, expanding common contractions, tokenizing, removing stopwords while preserving negations such as “not” and “never”, and preparing the resulting tokens for TF-IDF. Lemmatization was experimentally compared for the depression task; the final depression pipeline used the non-lemmatized representation based on the observed comparison.

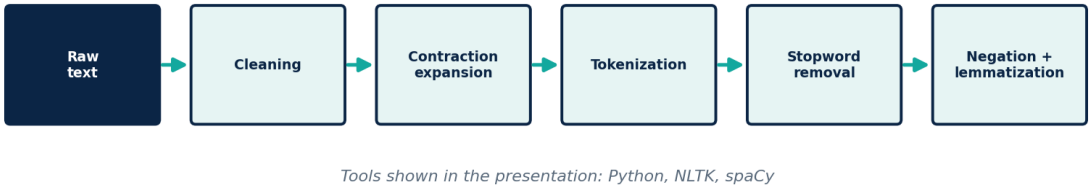


Figure 4. Text preprocessing workflow represented in the project materials.

4.3 TF-IDF Feature Extraction

TF-IDF converts text into sparse numeric feature vectors. Terms that are frequent in a document but less common across the corpus receive greater weight than ubiquitous terms. The vectorizer is fitted on training data only and then reused to transform validation and test data, keeping the feature vocabulary consistent.

4.4 Algorithms Compared

Algorithm	Role in experiment
Logistic Regression	Linear discriminative baseline.
Multinomial Naive Bayes	Probabilistic text-classification baseline.
Linear SVM	Maximum-margin linear classifier for sparse TF-IDF features.

Accuracy, precision, recall and F1-score were used for evaluation. Macro F1 is especially useful for the emotion task because it gives equal weight to each class.

CHAPTER 5: IMPLEMENTATION

5.1 Tools and Libraries

The project was implemented in Python using pandas and NumPy for data handling, NLTK for text-processing utilities, scikit-learn for TF-IDF and classification, Joblib for model persistence, and Streamlit for the web interface.

5.2 Model Training and Selection

Each task was trained and evaluated independently. Logistic Regression, Multinomial Naive Bayes and Linear SVM were compared on validation data. Linear SVM was selected for both tasks based on the reported comparison results. Final performance was then measured on the held-out test set.

5.3 Application and Saved Artifacts

The trained models and their fitted TF-IDF vectorizers are stored as Joblib files. At runtime, the Streamlit application loads these artifacts and performs inference without retraining. The two outputs are displayed separately.

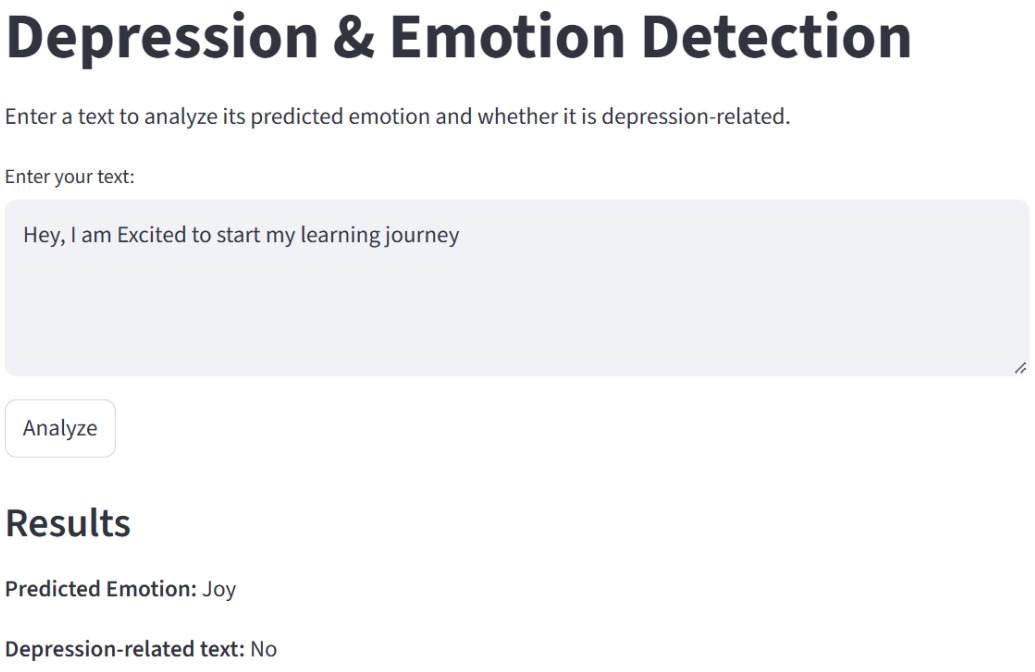


Figure 5. Web-application demo image included in the project presentation.

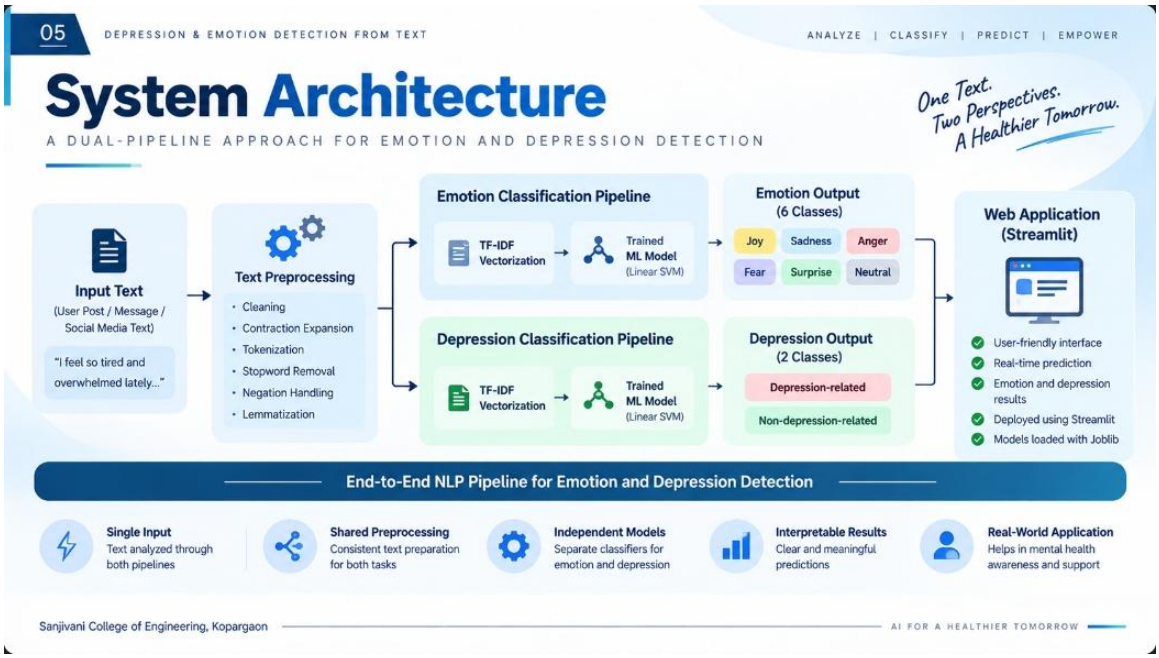


Figure 6. System architecture slide from the project presentation.



Figure 7. Team photograph

CHAPTER 6: RESULTS AND DISCUSSION

Linear SVM was the best-performing model in the reported comparisons for both tasks. The values below are the results produced during the project workflow.

6.1 Emotion Classification

Metric	Final test result
Accuracy	89.15%
Macro precision	0.86
Macro recall	0.84
Macro F1-score	0.85
Weighted F1-score	0.89

The test set contained 2,000 examples. Sadness and Joy were classified most reliably; Surprise was the weakest class by F1-score.

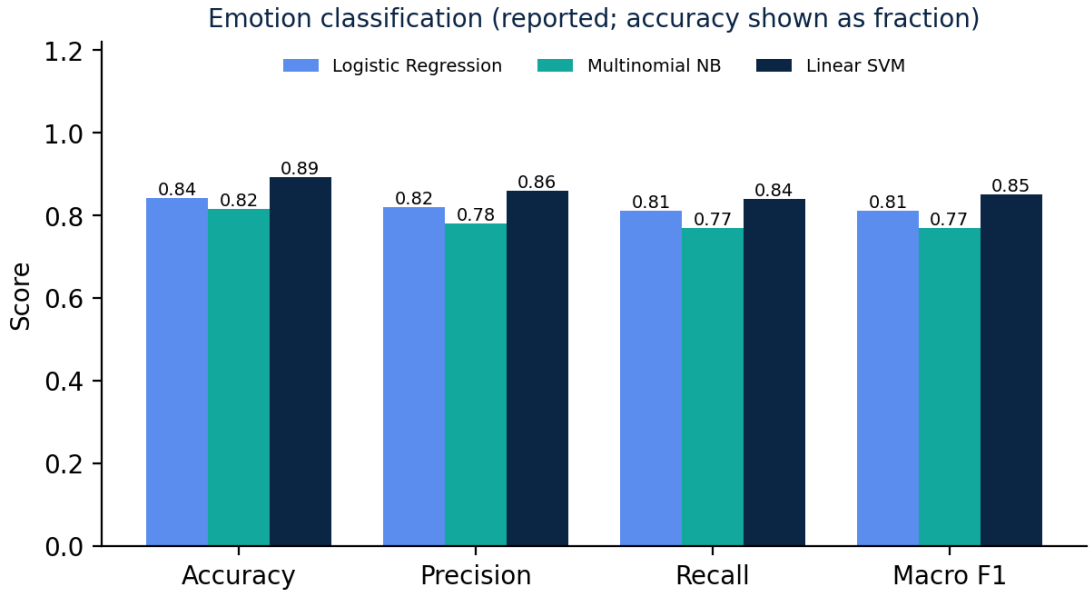


Figure 9. Emotion classifier comparison from the presentation.

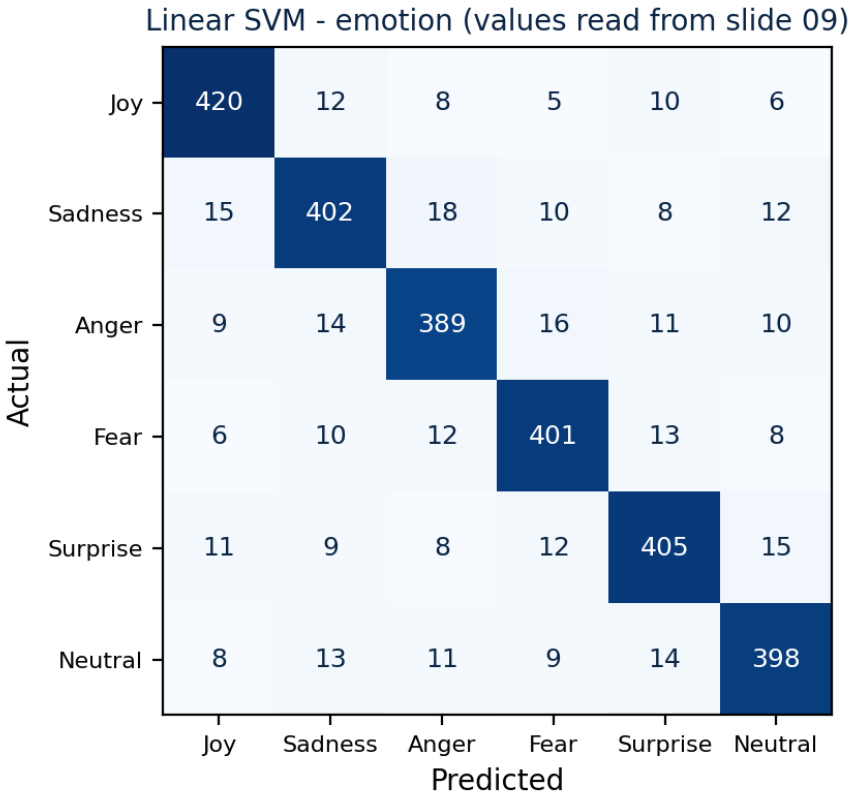


Figure 10. Emotion confusion matrix for Linear SVM (test set).

6.2 Depression-Related Text Classification

Metric	Final test result
Accuracy	92.70%
Macro F1-score	0.93
Weighted F1-score	0.93
Test samples	60,000

The confusion matrix showed 28,009 non-depression-related examples correctly classified, 27,609 depression-related examples correctly classified, 1,882 false positives and 2,500 false negatives. The result reflects classification against dataset labels, not a clinical assessment.

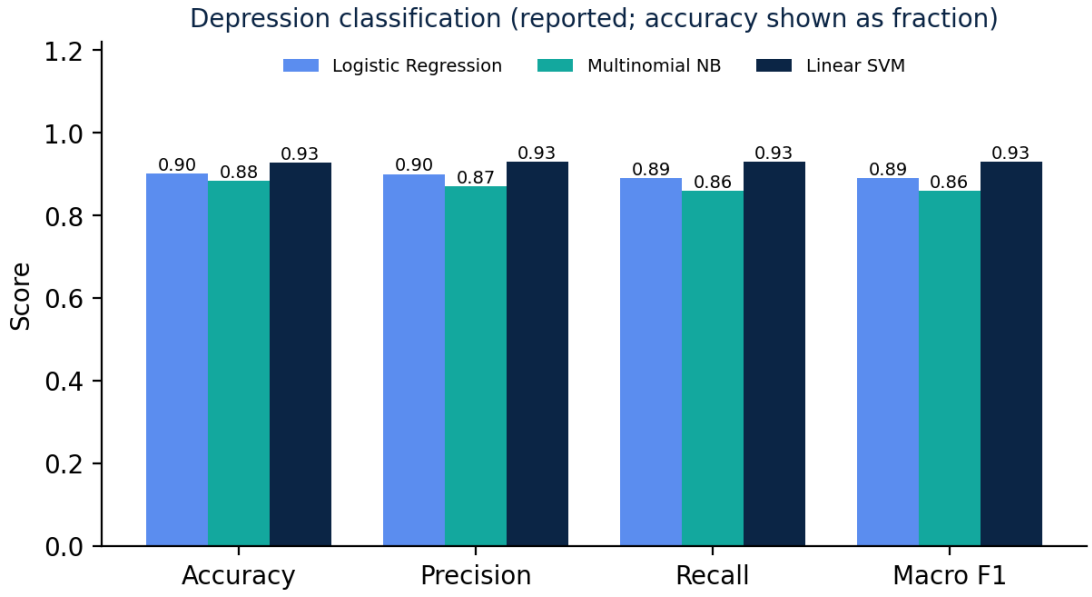


Figure 11. Depression-related classifier comparison from the presentation.

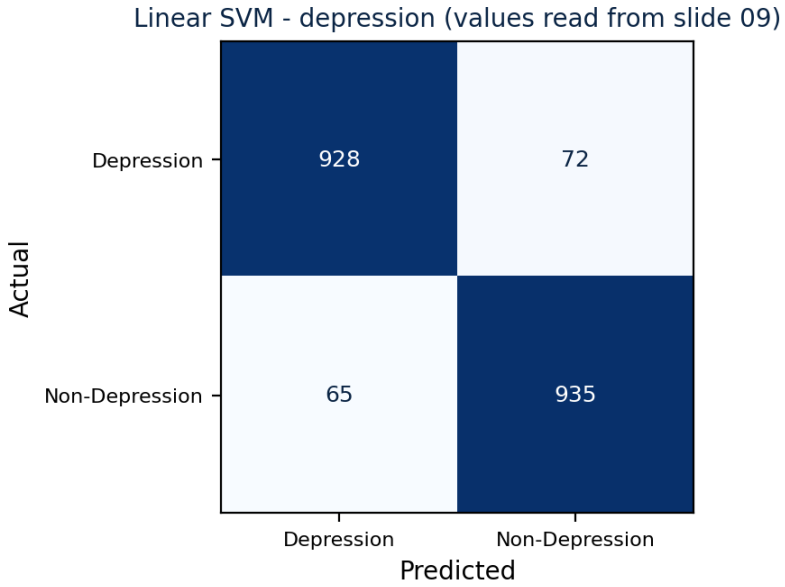


Figure 12. Depression-related text confusion matrix for Linear SVM (test set).

6.3 Interpretation and Limitations

The results indicate that TF-IDF with Linear SVM provides a strong classical baseline for these datasets. However, TF-IDF does not fully represent word order, context, sarcasm or subtle meaning. Dataset provenance, label quality and domain differences can affect performance. The model has not been clinically validated or externally validated on independent datasets.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The project implemented two independent NLP classification pipelines and deployed them through a Streamlit application. Linear SVM achieved 89.15% test accuracy with 0.85 macro F1 for six-class emotion classification, and 92.70% test accuracy with 0.93 macro F1 for binary depression-related text classification. The work demonstrates a complete classical ML workflow from preprocessing and TF-IDF to model comparison, evaluation and deployment.

7.2 Current Limitations

- The system supports English text and the labels represented in the training datasets.
- Sarcasm, complex negation, slang and context may cause errors.
- Performance may change on text from different platforms or populations.
- No clinical validation, depression-severity assessment or external dataset validation was performed.

7.3 Future Scope

- Evaluate on additional independent datasets and document dataset provenance.
- Explore contextual models such as BERT and compare them with the classical baseline.
- Improve handling of sarcasm, negation and code-mixed language.
- Add model-explanation features and strengthen privacy safeguards.

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